

Question:

The algorithm for calculating the Semester Grade Point Average (SGPA) and Cumulative Grade Point Average

(CGPA) for 'm' semesters is given below. Each semester has 'n' courses, where 'n' may vary between semesters.

Assume that the student has passed all courses. Prepare a test-case table covering valid, invalid, and boundary

inputs; write the C program based on the given algorithm; compile and execute it; and test it against the prepared test cases.

Grade point table:

grade	Grade point(GP)
O	10
A	9
B	8
C	7
D	6
E	5
P	4

The formulas used are:

$SGPA = \text{Semester Total Grade Credits} / \text{Semester Total Credits}$

$CGPA = \text{Cumulative Total Grade Credits} / \text{Cumulative Total Credits}$

2. Algorithm

1. Start.

2. Declare integer variables `global_id`, `m`, `n`, `semester`, `course`, `code`, `credit`, `GP`, `semester_TC`, `semester_TGC`, `cumulative_TC` and `cumulative_TGC`.
3. Declare `grade` as a character.
4. Declare `SGPA` and `CGPA` as floating-point variables.
5. Input `global_id`.
6. If `global_id <= 0`, display an error message and ask the user to enter it again.
7. Input number of semesters `m`.
8. If `m <= 0`, display an error message and ask again.
9. Initialize `cumulative_TC = 0` and `cumulative_TGC = 0`.
10. Repeat for every semester:
 - Set `semester_TC = 0` and `semester_TGC = 0`.
 - Input number of courses `n`.
 - If `n <= 0`, display an error and input again.
11. For every course:
 - Input course code.
 - If course code is not positive, input it again.
 - Input course credit.
 - If credit is not positive, input it again.
 - Input grade.
 - Accept only O, A, B, C, D, E, or P.
 - Use switch-case to find the grade point.
 - Calculate:
$$\text{semester_TGC} = \text{semester_TGC} + \text{GP} * \text{credit}$$

- Calculate:
 $\text{semester_TC} = \text{semester_TC} + \text{credit}$
- 12. Calculate:
 $\text{SGPA} = \text{semester_TGC} / \text{semester_TC}$
- 13. Display semester number and SGPA.
- 14. Add semester totals to cumulative totals.
- 15. After all semesters, calculate:
 $\text{CGPA} = \text{cumulative_TGC} / \text{cumulative_TC}$
- 16. Display Global ID, number of semesters, cumulative total credits and CGPA.
- 17. Stop.

C PROGRAM:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int global_id, m, n;
```

```
    int semester, course;
```

```
    int code, credit, GP;
```

```
    int semester_TC, semester_TGC;
```

```
    int cumulative_TC = 0, cumulative_TGC = 0;
```

```
    char grade;
```

```
    float SGPA, CGPA;
```

```
    /* Input and validation of Global ID */
```

```
do
{
    printf("Enter Global ID: ");
    scanf("%d", &global_id);

    if (global_id <= 0)
    {
        printf("Invalid Global ID. Enter a positive value.\n");
    }

} while (global_id <= 0);
```

```
/* Input and validation of number of semesters */
```

```
do
{
    printf("Enter number of semesters: ");
    scanf("%d", &m);

    if (m <= 0)
    {
        printf("Invalid number of semesters. Enter a positive value.\n");
    }

} while (m <= 0);
```

```
/* Loop for each semester */  
for (semester = 1; semester <= m; semester++)  
{  
    semester_TC = 0;  
    semester_TGC = 0;  
  
    printf("\nSemester %d\n", semester);  
  
    /* Input and validation of number of courses */  
    do  
    {  
        printf("Enter number of courses: ");  
        scanf("%d", &n);  
  
        if (n <= 0)  
        {  
            printf("Invalid number of courses. Enter a positive value.\n");  
        }  
    } while (n <= 0);  
  
    /* Loop for each course */
```

```
for (course = 1; course <= n; course++)
{
    printf("\nCourse %d\n", course);

    /* Input course code */
    do
    {
        printf("Enter course code: ");
        scanf("%d", &code);

        if (code <= 0)
        {
            printf("Invalid course code. Enter a positive value.\n");
        }

    } while (code <= 0);

    /* Input course credit */
    do
    {
        printf("Enter course credit: ");
        scanf("%d", &credit);

        if (credit <= 0)
```

```

    {
        printf("Invalid credit. Enter a positive value.\n");
    }

} while (credit <= 0);


/* Input grade */
do
{
    printf("Enter grade (O/A/B/C/D/E/P): ");
    scanf(" %c", &grade);

    if (grade != 'O' && grade != 'A' &&
        grade != 'B' && grade != 'C' &&
        grade != 'D' && grade != 'E' &&
        grade != 'P')
    {
        printf("Invalid grade. Enter O, A, B, C, D, E or P.\n");
    }

} while (grade != 'O' && grade != 'A' &&
        grade != 'B' && grade != 'C' &&
        grade != 'D' && grade != 'E' &&
        grade != 'P');

```

```
/* Convert grade into grade point */  
switch (grade)  
{  
    case 'O':  
        GP = 10;  
        break;  
  
    case 'A':  
        GP = 9;  
        break;  
  
    case 'B':  
        GP = 8;  
        break;  
  
    case 'C':  
        GP = 7;  
        break;  
  
    case 'D':  
        GP = 6;  
        break;
```

```
        case 'E':  
            GP = 5;  
            break;  
  
        case 'P':  
            GP = 4;  
            break;  
    }  
  
    /* Calculate semester totals */  
    semester_TGC = semester_TGC + (GP * credit);  
    semester_TC = semester_TC + credit;  
}  
  
/* Calculate SGPA */  
SGPA = (float)semester_TGC / semester_TC;  
  
printf("\nSGPA of Semester %d = %.2f\n",  
    semester, SGPA);  
  
/* Add semester values to cumulative values */  
cumulative_TGC = cumulative_TGC + semester_TGC;
```

```

        cumulative_TC = cumulative_TC + semester_TC;
    }

    /* Calculate CGPA */

    CGPA = (float)cumulative_TGC / cumulative_TC;


    /* Display final result */

    printf("\n----- Final Result ----- \n");
    printf("Global ID = %d\n", global_id);
    printf("Number of Semesters = %d\n", m);
    printf("Cumulative Total Credits = %d\n", cumulative_TC);
    printf("CGPA = %.2f\n", CGPA);

    return 0;
}

```

Test case table:

Test case	Input	Expected Result
Valid	Global ID = 101, m = 1, n = 2; Course 1: code=111, credit=4, grade=O; Course 2: code=112, credit=3, grade=A	SGPA = (40+27)/7 = 9.57, CGPA = 9.57
Valid multiple semesters	Global ID = 101, m = 2 with valid courses in both semesters	Program calculates and displays SGPA of both semesters and final CGPA

Invalid Global ID	Global ID = -10, then 101	Error for -10; accepts 101
Invalid semesters	m = 0, then 2	Error for 0; accepts 2
Invalid courses	n = -2, then 3	Error for -2; accepts 3
Invalid course code	code = 0, then 101	Error for 0; accepts 101
Invalid credit	credit = -4, then 4	Error for -4; accepts 4
Invalid grade	grade = X, then A	Error for X; accepts A
Lower boundary	Global ID=1, m=1, n=1, code=1, credit=1, grade=P	SGPA = 4.00, CGPA = 4.00
Highest grade	One course, credit=1, grade=O	SGPA = 10.00, CGPA = 10.00

Execution and compilation:

```

(kali㉿kali)-[~]
$ vi h19.c

(kali㉿kali)-[~]
$ gcc h19.c -o h19

(kali㉿kali)-[~]
$ ./h19
Enter Global ID: 101
Enter number of semesters: 1

Semester 1
Enter number of courses: 1

Course 1
Enter course code: 111
Enter course credit: 4
Enter grade (O/A/B/C/D/E/P): o
Invalid grade. Enter O, A, B, C, D, E or P.
Enter grade (O/A/B/C/D/E/P): O

SGPA of Semester 1 = 10.00

—— Final Result ——
Global ID = 101
Number of Semesters = 1
Cumulative Total Credits = 4
CGPA = 10.00

(kali㉿kali)-[~]

```